LAB EXPERIMENT
ITA0506-COMPUTER VISION

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EXPNO: 33. Morphological Gradient Technique

Aim

To perform morphological gradient on an image using OpenCV.

Algorithm

1. Import OpenCV.
2. Read BEN 10.png.
3. Convert the image to grayscale.
4. Create a morphological kernel.
5. Apply morphological gradient using cv2.morphologyEx().
6. Display and save the result.

CODE:

EXP-33.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

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```python
import cv2
import numpy as np

# Read image
img = cv2.imread("BEN 10.png")

if img is None:
    print("Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply morphological gradient
gradient = cv2.morphologyEx(
    gray,
    cv2.MORPH_GRADIENT,
    kernel
)

# Display result
cv2.imshow("Original Image", gray)
cv2.imshow("Morphological Gradient", gradient)

# Save result
cv2.imwrite("morphological_gradient.png", gradient)

cv2.waitKey(0)
cv2.destroyAllWindows()

print("Morphological gradient completed successfully!")
```

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OUTPUT :

Result

The morphological gradient is successfully obtained, highlighting the boundaries of objects in the image.